

Problem 1 (02IMO2)

Angles directed by default

We show that $\overline{CJ}$ bisects $\angle ECF$.

$$\angle ECJ = \angle JCF \Leftrightarrow \angle EOA = \angle AOF$$

Note that $\angle EOA = \angle AOF = 60$ since FE is the perpendicular bisector of OA so we are done.

We show that $\overline{EJ}$ bisects $\angle FEC$. Let $2x = \angle AOB$.

$$\angle FEC = \angle AEB - \angle AEF + \angle BEC = x - 30 + 90 = x + 60$$

We now find $\angle FEJ$. Note that $CA \parallel DO$ since $\angle BOD = \angle BCA = x$. Similarly, $DA \parallel OJ$ as we can angle chase to get $\angle DAO = 90 - \frac{x}{2} = 180 - 2x - \angle COJ = \angle JOA$. Therefore, $AJOD$ is a parallelogram and M is the midpoint of DJ .

Since $ME = MF$ as well, we have $\angle FEJ = \angle EFD = \angle EFA + \angle AFD = 30 + \frac{x}{2} = \frac{1}{2}(\angle FEC)$ as desired.

Problem 2 (GGG44)

WLOG let $AB < AC$. Note that D, O, X are collinear if and only if $DO \perp PQ$.

We first show that there is at most one point D on BC which results in $DO \perp PQ$, then identify the point and prove it leads to $AX \parallel BC$ to suffice.

Let BC be the x -axis and let A have positive y coordinates. Let O have x -coordinate zero and B negative x -coordinate. Note that if D has nonnegative x -coordinate, then the slope of DO is undefined or negative. However, since $\angle C < \angle B$ and $DC < DB$, the y -coordinate of Q is less than that of P , meaning the slope of PQ is strictly negative. Therefore, it is impossible for $DO \perp PQ$ in this case.

We now slide D from B to the midpoint of BC and show that there is only one point where $DO \perp PQ$ may be satisfied.

Since P, Q lie on the perpendicular bisector of BD, CD respectively, the difference in the x -coordinate of P and Q is constant at $\frac{a}{2}$. As D travels from B to the midpoint of BC , the y -coordinate of P increases and the y -coordinate of Q decreases, meaning the slope s_1 of PQ will decrease. The slope s_2 of DO increases and remains positive. $DO \perp PQ$ if and only if $s_1 s_2 = -1$, which requires $s_2 < 0$. After passing the threshold where $s_2 < 0$, as D continues

traveling, $|s_2|$ and $|s_1|$ are both strictly increasing. Hence, there is only one point D where both $|s_1 s_2| = 1$ and $s_1 s_2 < 0$ is possible as desired.

Let A' be the reflection of A over the perpendicular bisector of BC . We show that, if $D = A'O \cap BC$, we have $PQ \perp DO$ and $X = A'$. This suffices to show both directions since $AX = AA' \parallel BC$ and there can be no other values of X .

Let the line perpendicular to BC through D intersect AC at Q_2 . We show that CDQ_2A' is a cyclic quadrilateral.

$$\angle XDQ_2 = 90 - \angle CDX$$

By reflection over the perpendicular bisector of BC ,

$$= 90 - \angle(AO, BC) = 90 - (90 - \angle B + \angle C) = \angle B - \angle C$$

As desired,

$$\angle XCQ_2 = \angle XCA = \angle CBA - \angle XBC = \angle CBA - \angle ACB = \angle B - \angle C$$

Since CDQ_2A' is cyclic, we see that the midpoint of CQ_2 , Q , is the center of this circle. Therefore,

$$QC = QD = QA'$$

Q lies on the perpendicular bisector of $A'D$. Analogously, P lies on the perpendicular bisector as well meaning $PQ \perp DA' = DO$ and $X = A'$ as desired.

5 clubs; 2.5 hours

Problem 4 (Z57F7BDO)

Let O, O_1, O_2 be the centers of $\Omega, \gamma_1, \gamma_2$ respectively. Let QT intersect Ω again at S .

By a homothety at Q , we have $O_2T \parallel OS$. Since $O_2 - T - O_1, TO_1 \parallel OS$.

It suffices to show $SP \perp PT$. Extend TO_1 to meet γ_1 again at T' . By a homothety at $P, T'O_1 \parallel OS \Leftrightarrow P - T' - S$. $90 = \angle T'PT = \angle SPT$ as desired.

Problem 5 (07SLG2)

Claim: The reflections of B over MX and C over TM are the same.

We employ coordinates.

Let TM be the x -axis and MX be the y -axis. Note $M = (0, 0)$ is the midpoint of BC , so let,

$$B = (m, n); C = (-m, -n)$$

Note that the reflection of C over $\overline{TM}$ and the reflection of B over $\overline{MX}$ are both $(m, -n)$. $\square$

Let the two reflections both go to Y . Since B reflects through MX to Y , $XB = XY$ meaning (TBY) is centered at X . Furthermore, $\angle BXM = \angle YXM$ by the reflection. Therefore,

$$\begin{aligned}\angle BXY &= 2\angle BTY \Leftrightarrow \angle BXM = \angle BTY \Leftrightarrow \angle BAM = \angle BTY \\ \Leftrightarrow \angle BAM &= \angle BTC - 2\angle YTC \Leftrightarrow \angle BAM = \angle BTC - 2\angle MTC \\ \Leftrightarrow \angle BAM &= \angle MTB - \angle CTM\end{aligned}$$

Since $\angle BAM$ does not depend on X , we are done.

11 clubs; 4 hours

Problem 7 (18IMO1)

Let the perpendicular bisector of BD intersect major arc AB at F_1 and perpendicular bisector of CE intersect major arc AC at G_1 .

Claim: $FF_1 = GG_1$.

It suffices to show lines FF_1 and GG_1 are equidistant from O the circumcircle. Indeed, if $AD = AE = d$, then observe O is $\frac{d}{2}$ from FF_1 since the midpoint M_c of AB is $\frac{d}{2}$ from FF_1 . Similarly, O is $\frac{d}{2}$ from GG_1 as desired. $\square$

With the claim proven, let $P = FF_1 \cap GG_1$. We have $PF = PG$.

Let l_1 be the angle bisector of $\angle FPG$ and be the l_2 - the angle bisector of $\angle BAC$. Since $FP \perp AB$ and $GP \perp AC$, $l_1 \parallel l_2$.

Note $l_1 \perp FG$ by PFG being isosceles and $l_2 \perp DE$ by ADE being isosceles so we are done.

Problem 8 (18CHNTST21)

Directed angles in use by default.

Let the perpendicular bisector of $\overline{BC}$ intersect the internal angle bisector of $\angle B$ at X and that of $\angle C$ at Y . Let I be the incenter. We claim that P lies on (XYI) .

Let D' be the reflection of D over the midpoint of AB . Observe that, since X lies on the perpendicular and angle bisector, and since $BD' = BF$, we have congruent triangle conditions giving us,

$$\angle XDC = \angle XD'B = \angle XFB = \angle XFC$$

Therefore, $(BDXF)$ is cyclic. Symmetrically, $(CDYE)$ is cyclic.

Now angle chase to get,

$$\begin{aligned}\angle XGY &= \angle XGD + \angle DGY \\ &= \angle XBD + \angle DCY = \angle IBC + \angle BCI = \angle BIC \\ &= \angle XIY\end{aligned}$$

As desired.

21 clubs; 5.5 hours

Problem 9 (07IMO2)

Angles directed by default.

Given $EF = EG = EC$, if we apply a homothety at C of scale $\frac{1}{2}$, F goes to the base of the altitude from E to CD , G goes to the base of the altitude from E to BC , and A goes to the midpoint M of BD . P_1, P_2 , and M are collinear.

By the Simson line, P_1, P_2 , and the base of the altitude from E to BD are collinear as well. The base of the altitude from E to BD is M , meaning $BE = DE$.

$$\begin{aligned}BE = DE &\Leftrightarrow \angle BCE = \angle DCE \\ &\Leftrightarrow \angle GCE = \angle FCE\end{aligned}$$

Since (FEG) is centered at C , $FC = FG$.

$$\begin{aligned}FC = FG &\Leftrightarrow \angle CFG = \angle FGC \\ &\Leftrightarrow \angle DFA = \angle AGB\end{aligned}$$

By parallelogram conditions,

$$\Leftrightarrow \angle FAB = \angle GAD$$

Hence, $\overline{AGF}$ is the internal angle bisector of $\angle BAD$.

Problem 11 (15IMO4)

WLOG, let $AB \leq AC$.

Lemma: $\angle CEG - \angle FDB = \angle B - \angle C$

Proof: Note $AF = AG$ implies AO bisects $\angle FAG$. Therefore,

$$\angle GAM - \angle FAM = 2\angle MAO = 2(\angle B - \angle C)$$

Let M be the midpoint of minor arc DE .

Since $\angle DAM = \angle EAM$,

$$\angle GAE - \angle FAD = 2(\angle B - \angle C)$$

Since AFD, AEG are isosceles,

$$\angle ADF - \angle AEG = \angle B - \angle C$$

Using $\angle ADE = \angle AED$, we get $\angle CEG - \angle FDB = \angle B - \angle C$ as desired. $\square$

We angle chase. Let $DF \cap AO = P_1; GE \cap AO = P_2; AO \cap BC = P$.

$$\angle APC = 90 - C - B; \angle APB = 90 + C - B$$

Therefore,

$$\begin{aligned} \angle AP_2G - \angle AP_1F &= \angle EP_2P - \angle PP_1D \\ &= (180 - \angle PEP_2 - \angle P_2PE) - (180 - \angle P_1DP - \angle DPP_1) \\ &= \angle B - \angle C \end{aligned}$$

Observe that $AG = AF$ and AO being diameter implies $\angle GAP_2 = \angle FAP_1$. Angle chasing on triangles AFP_1, AP_2G , we get,

$$\angle P_1FA - \angle P_2GA = \angle B - \angle C$$

Since $\angle P_1FK = \angle DFK = \angle DBK = \angle B$ and $\angle P_2CL = \angle C$, we get,

$$\angle KFA = \angle LGA$$

Let $FK \cap AO = Q_1; GL \cap AO = Q_2$.

Note that AQ_1F and AQ_2G are congruent by ASA congruency so $AQ_1 = AQ_2$. Since Q_1, Q_2 lie in (ABC) , $Q_1 = Q_2$ as desired.

Problem 12 (17IMO4)

Directed angles in use.

Let K_1 be the intersection of the tangent to Γ at T with Ω closer to T . It suffices to show that A, J, K_1 are collinear since that would imply $K_1 = K$.

Let K' be the reflection of K_1 over S and A' be the reflection of A over S .

Lemma: A, K', T are collinear.

Since $RK \parallel TK'$ by the reflection,

$$\angle AK'S = \angle ARS = \angle RK_1S = \angle TK'S \quad \square$$

With the claim proven, we now get $RK_1 \parallel AT$.

$$\angle SRK = \angle STA \Leftrightarrow \angle SJK = \angle SJA$$

As desired.

Problem 13 (12SLG4)

Let N be the base of the altitude from A to BC . $AN \parallel AX$ and AD bisects $\angle NAX$ means,

$$\angle ADX = \angle DAX \Leftrightarrow AX = DX$$

Let P be the base of the altitude from X to AM . Since AXD is isosceles, P is the midpoint of AD . By Power of a Point,

$$DA \cdot DM = DB \cdot DC$$

Note that Y is the reflection of D over M .

$$\Leftrightarrow DP \cdot DY = DB \cdot DC$$

Therefore, $(BPCY)$ is cyclic. It suffices to show $(BPXC)$ is cyclic.

Let $(BPCY)$ intersect the line perpendicular to BC at D at a point X' on the same side of BC as A . The projection P_1 of the circumcenter of $(BPCY)$ onto BC is the midpoint of the projections of X' and Y onto BC . Since $X'Y$ is not parallel to BC as they lie on different sides of BC , X' and Y are antipodes.

$$\angle X'PD = 90$$

We also have $\angle XPD = 90$ from the isosceles so $X' = X$ and we are done.

39 clubs; 10 hours

Problem 15 (Z2CD5C2A)

Directed angles in use

Lemma: $(AXE), (AYF)$ are tangent to EF .

Proof: Note,

$$\angle AXE = \angle AXD = \angle ACD = \angle AEF$$

Therefore, (AXE) is tangent to EF . Analogously, (AYF) is tangent to EF .

Let $(AEX), (AYF)$ intersect again at Q ; $(AYF), (DYX)$ intersect again at G . Let D'_1 be where (YDX) intersects BC again. Our goal is to show $D'_1 = D_1$.

Claim 1: G, F, D'_1 are collinear.

Proof: Indeed,

$$\angle GD'_1D = \angle GYD = \angle GYF = \angle GFE$$

Since $FE \parallel D'_1D$, the claim is proven. $\square$

Claim 2: $(GFEX)$ is cyclic.

Proof:

$$\angle GFE = \angle GYF = \angle GYD = \angle GXD = \angle GXE \quad \square$$

Let $P = GF \cap XE$. From $(GFEX)$ being cyclic, P lies on the radical axis AM of (AYF) and (AXE) . Therefore, P, M, A are collinear.

The midpoint M_1 of EF satisfies $M_1E^2 = M_1F^2$ so it is also on the radical axis.

By a homothety at P , the midpoint M_2 of DD'_1 is also on the radical axis.

Since A, M_1, M_2 are collinear, M_2 is also the midpoint of BC . Hence, $D'_1 = D_1$ as desired.

Problem 16 (22APMO2)

We claim the fixed point is C' the reflection of C over B .

Let F' be where $\overline{C'E}$ intersects (BDE) again. Showing $(AF'DC)$ is cyclic suffices as it would imply $F' = F$, meaning $C' - F - E$ as desired.

Claim: The C'' the reflection of C' across E lies on (ACD)

Proof: We show that $AC''D$ is in fact an isosceles trapezoid. Let C_3 be the base of the altitude from C'' to BC and E_3 be the base of the altitude from E to BC . We have,

$$AB = 2 \cdot EE_3 = C''C_3$$

$$CC_3 = BC - AC'' = BC' - C'D = DB$$

Therefore, triangles ABD and $C''C_3C$ are congruent as desired. $\square$

We can now apply Power of a Point.

$$C'D \cdot C'B = C'F' \cdot C'E$$

Doubling both sides,

$$\Leftrightarrow C'D \cdot C'C = C'F' \cdot C'C''$$

$(DF'C''C)$ is cyclic so $(DF'AC)$ is cyclic as desired.

47 clubs; 12 hours

Problem 18 (23ESLG7)

Claim: Q is the excentre of both $\triangle PF_1B$ and $\triangle PF_2A$.

Proof: We first show that the two triangles have the same excentre. First note that their excentres E_1, E_2 both lie on the internal angle bisector of $\angle F_1PF_2$. Furthermore, since,

$$PF_1 + F_1B + BP = 2PF_1 + 2PF_2 = PF_2 + F_2A + AP$$

The two triangles have the same semi-perimeter and the projections of E_1, E_2 onto PB are the same point. Therefore, $E_1 = E_2$.

With $E_1 = E_2$ shown, observe BQ is the external angle bisector of $\angle F_1BF_2$ and AQ external angle bisector of $\angle F_1AF_2$. Q is collinear with BE_1 and AE_2 so $Q = E_1 = E_2$. $\square$

We now show M is where the internal angle bisector of PF_1F_2 intersects (PF_1F_2) . From the excentre condition, Q lies on the internal angle bisector. Let Q' be the projection of Q onto AP and M' be that of M . We show that $PM' = \frac{1}{2}PQ'$ to suffice.

Note that PQ' is the semi-perimeter of PF_2A , which is $\frac{1}{2}(2PF_1 + 2PF_2) = PF_1 + PF_2$. It is well known that PM' has length $\frac{1}{2}(PF_1 + PF_2)$ so we are done (e.g. by observing M is midpoint of incenter, P -excentres of PF_1F_2).

Problem 19 (18TSTST5)

Directed angles in use.

Let M_B, M_C be midpoints of AC, AB respectively and O be center of (ABC) .

Claim: (OAE_1F_1) is cyclic.

Since $\angle OPE_1 = \angle OM_C A = \angle OM_B A = 90$, $(OM_C PF_1)$ and $(OM_C AM_B)$ are cyclic. We angle chase,

$$\angle OF_1 E_1 = \angle OM_C P = \angle OM_C M_B = \angle OAM_B = \angle OAE_1 \quad \square$$

Analogously we may prove $(OAE_2 F_2)$ is cyclic.

Since OA is radical axis, we have proven the line through center of circles is perpendicular to OA and hence parallel to tangent at A .

Now note there is a spiral similarity at O sending $E_1 F_1$ to $E_2 F_2$ from $(OAE_1 F_1)$ and $(OAE_2 F_2)$ being cyclic. There is also a spiral similarity sending PE_1 to QE_2 from $(OM_B PE_1)$ and $(OM_B QE_2)$ being cyclic. Therefore, $OF_1 PE_1 \sim OE_2 QF_2$.

Since $OQ = OP$, the two are congruent and $E_1 F_1 = E_2 F_2$.

57 clubs; 15 hours

Problem 22 (99IMO5)

Let O_1, O_2 be centers of ω_1, ω_2 , and let T be where segment $O_1 O_2$ intersects ω_2 . Let MO_2 intersect ω again at X . And XO intersect ω again at Y . Let P be the midpoint of AB .

Since O_2 lies on ω_1 , homothety gives $O_1 O_2 \parallel OW$. Therefore, a homothety at N taking O_2 to O would take T to Y so $Y - T - N$.

Note that $MT O_2 N$ is cyclic because,

$$\angle TO_2 M = \angle YXM = \angle YNM = \angle TNM$$

Claim: $MOPN$ is cyclic.

Proof: Let K be radical center of $\omega_1, \omega_2, \omega$. Since KM, KN are tangents and $K - A - B$,

$$\angle OMK = \angle ONK = \angle OPK = 90$$

We have $(OPNKM)$ cyclic as desired. $\square$

We now angle chase,

$$\angle OPM = \angle ONM = \angle MTO_2 = \angle O_1TM$$

Therefore, $M - T - P$.

The homothety sending T to P sends O_2 to X , so it sends ω_1 to ω . Therefore, $C - D - T$. Since $CD \parallel AB \perp TO_2$ by homothety, we have CD is the tangent as desired.

Problem 23 (19EGMO4)

A is on radical axis of (BPI) , (CQI) so $(BPQC)$ is cyclic. Angle chase to get,

$$\begin{aligned}\angle BPI &= 180 - \angle PBI - \angle PIB = 180 - \left(\frac{\angle B}{2}\right) - (\angle AIB - \angle AIP) \\ &= 180 - \left(\frac{\angle B}{2}\right) - \left(\left(90 + \angle \frac{C}{2}\right) - \left(\frac{\angle B}{2}\right)\right) = 90 - \angle \frac{C}{2}\end{aligned}$$

Since $\angle BPQ = 180 - \angle C$, we find PI bisects $\angle BPQ$. Analogously, QI bisects $\angle CQP$. Therefore, I is the A -excentre APQ as desired.

Sixseven clubs; 16.5 hours

Problem 24 (21SLG4)

Let DT meet AC at H and $(ABCD)$ again at W . Let $CD \cap BW = G$ and $AD \cap BW = I$. By Pascal's on $ABWDDC$ and $CBWDDA$ respectively, $E - H - G$ and $I - H - F$. Furthermore, from $ET \parallel DG$,

$$\frac{EH}{HG} = \frac{TH}{HD}$$

We may analogously get $\frac{FH}{HI} = \frac{TH}{HD}$. Therefore, there is a homothety h at H sending EDF to GWI . Note that $BW \parallel DD$ by the homothety, so BDW is isosceles. Since DTK is isosceles as well, $TK \parallel BD$.

Now note since $ET \parallel DG$ and $TF \parallel AD$, the same h homothety sends TEF to DGI .

As a result, since $TK \parallel BD$, h sends K to B and we are done.